

Umar Mohmed

B.Tech (2022–26)
Computer Science and Engineering (AI & ML)
Institute of Aeronautical Engineering, Hyderabad

+91-9392466218
umarmd0507@gmail.com
github.com/07-Umar
leetcode.com/u/Umar_7
linkedin.com/in/umarmohmed

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. CSE (AI & ML)	Institute of Aeronautical Engineering	9.16	2022–2026 (Expected)
Senior Secondary	TSBIE Board	96.2%	2019–2021
Secondary	SSC Board	10/10	2018–2019

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Machine Learning

EXPERIENCE

Research Intern

May 2024 – Aug 2024

Institute of Aeronautical Engineering

Hyderabad, Telangana

- Developed a highly accurate Machine Learning-based **Network Intrusion Detection System (NIDS)** using benchmark datasets (KDD Cup 99, NSL-KDD), leveraging the **Random Forest Classifier** to achieve **99.81% accuracy** in detecting and classifying network anomalies.
- Optimized system performance through **feature selection, SMOTE balancing, and hyperparameter tuning, enhancing real-time detection accuracy, scalability, and fault tolerance.**
- Tech Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn.

PROJECTS

Personality Detection Using XLM-RoBERTa and Whisper

June 2025 – Sept 2025

Multimodal Machine Learning & NLP Project (B.Tech Final Year)

- Developed a real-time **MBTI personality detection system** by integrating **OpenAI Whisper (ASR)** with a fine-tuned **XLM-RoBERTa** transformer model, achieving **85.9% classification accuracy** across **16 personality types**.
- Built an end-to-end **multimodal NLP pipeline** with **audio transcription, text preprocessing, hybrid resampling, transformer-based classification, and FastAPI deployment**, enabling scalable **AI-driven psychological analytics for HR, education, and digital wellness applications.**
- Tech Stack:** Python, Hugging Face Transformers, XLM-RoBERTa, OpenAI Whisper, PyTorch, FastAPI.

AI Smartboard

May 2025 – Aug 2025

Generative AI-Powered Interactive Whiteboard Application

GitHub | Live Link

- Developed an **AI-powered whiteboard** solving handwritten equations, problems in real time with Gemini API.
- Added **step-by-step explanations, PDF export, and multi-page support**; deployed via Vercel + Render.
- Tech Stack:** Python, FastAPI, React, TypeScript, Canvas API, Google Gemini API, Tailwind CSS, Vercel, Render.

TECHNICAL SKILLS

- Programming:** Python, Java, SQL
- AI & ML:** Machine Learning, Deep Learning, NLP, GenAI, Model Training and Evaluation, Feature Engineering
- Generative AI & NLP:** Prompt Engineering, LLM APIs (Gemini), Hugging Face Transformers, XLM-RoBERTa
- Data Handling & Analysis:** Data Preprocessing, Exploratory Data Analysis (EDA), NumPy, Pandas
- ML Tools & Libraries:** Scikit-learn, Matplotlib, Seaborn, PyTorch, Whisper (Speech-to-Text)
- Systems & Tools:** Git, GitHub, Linux/Unix
- Soft Skills:** Problem Solving, Analytical Thinking, Communication, Team Collaboration, Continuous Learning

CERTIFICATIONS

- Java Full Stack Certification Program – **Wipro TalentNext** Oct. 2025
- Amazon ML Summer School 2025 – **Amazon** Sept. 2025

PUBLICATIONS

- Mohmed Umar, et al. “**Personality Detection Using XLM-Roberta and Whisper.**” *IEEE Int. Conference*, 2025. [↗](#)
- Mohmed Umar, et al. “**Network Intrusion Detection System Using Learning Algorithms.**” *Springer Nature*, 2025. [↗](#)

ACHIEVEMENTS

- Completed **Amazon ML Summer School 2025 (Top 5% of 60,000+ applicants)**, an advanced ML training program conducted by Amazon scientists.
- Awarded **Academic Excellence Medals at IARE Spandana — Gold (3rd and 5th semesters) and Silver (1st semester)** for consistent academic performance.
- Secured **1st place in Tech Havoc (Consortium '23)** for innovative problem-solving and technical implementation.
- Solved 300+ Data Structures and Algorithms** problems across LeetCode and GeeksforGeeks; achieved **LeetCode** rating of **1728 (Top 11% globally)**.
- Published 2 research papers** in ML and NLP domains (**Springer Nature, IEEE Conference, 2025**).
- Mentored CBSE Class 10 students** through **personalized home tuition**, helping students achieve **87% and 89% academic scores.**